Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Surds & Indices

MARKSCHEME

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1. $5\sqrt{2}+3\sqrt{2}=8\sqrt{2}$ **M1A1** [3]
2. $5\sqrt{3}-2\sqrt{3}=3\sqrt{3}$ **M1A1** [3]
3. $4\sqrt{3}+6\sqrt{3}=10\sqrt{3}$ **M1A1** [3]
4. $\frac{\sqrt{5}}{5}$ **M1A1** [2]
5. $\frac{3\sqrt{7}}{7}$ **M1A1** [2]
6. Multiply by $\sqrt{3}+1$: $\sqrt{3}+1$ **M1A1** [3]
7. $\frac{5(\sqrt{5}-2)}{1}=5\sqrt{5}-10$ **M1A1** [3]
8. $\frac{\sqrt{15}+\sqrt{6}}{3}$ **M1A1** [4]
9. 4 **B1** [2]
10. 9 **B1** [2]
11. $\frac{1}{8}$ **M1A1** [3]
12. 16 **M1A1** [3]
13. $1+\sqrt{3}$ **M1A1** [3]
14. $20-4\sqrt{5}$ **M1A1** [3]
15. 18 **M1A1** [3]
16. $7+4\sqrt{2}$ **M1A1** [4]
17. $\frac{11+5\sqrt{5}}{4}$ **M1A1** [4]
18. $x=8$ **M1A1** [3]
19. $9^x=3^{2x}$, $27^{x-1}=3^{3x-3}$; $2x=3x-3\Rightarrow x=3$ **M1A1** [4]
20. $2^{2x+2}=2^{3(2x-1)}\Rightarrow x=\frac{5}{4}$ **M1A1** [4]
21. 5 **M1A1** [3]
22. $2\sqrt{3}$ **M1A1** [4]
23. Add/sub: $2\sqrt{a}=6\Rightarrow a=9$; $b=4$ **M1A1** [5]
24. 2 **M1A1** [3]
25. Square: $2x+1=(x-1)^2$; $x=4$ (reject $x=0$) **M1A1** [5]